

Smart Dust Sentinel Web

Edge-collaborative threat detection
that keeps grain-sized covert motes
invisible and autonomous

By MajorKali



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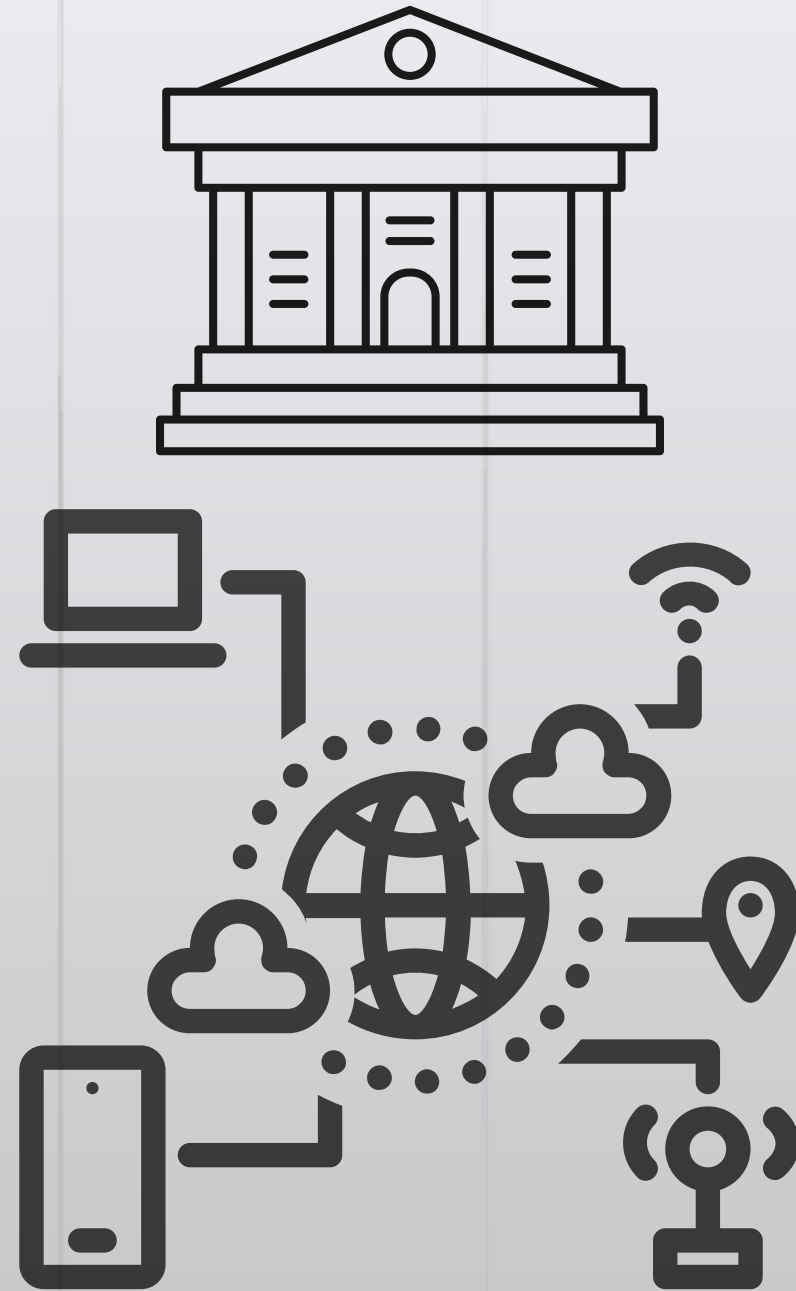
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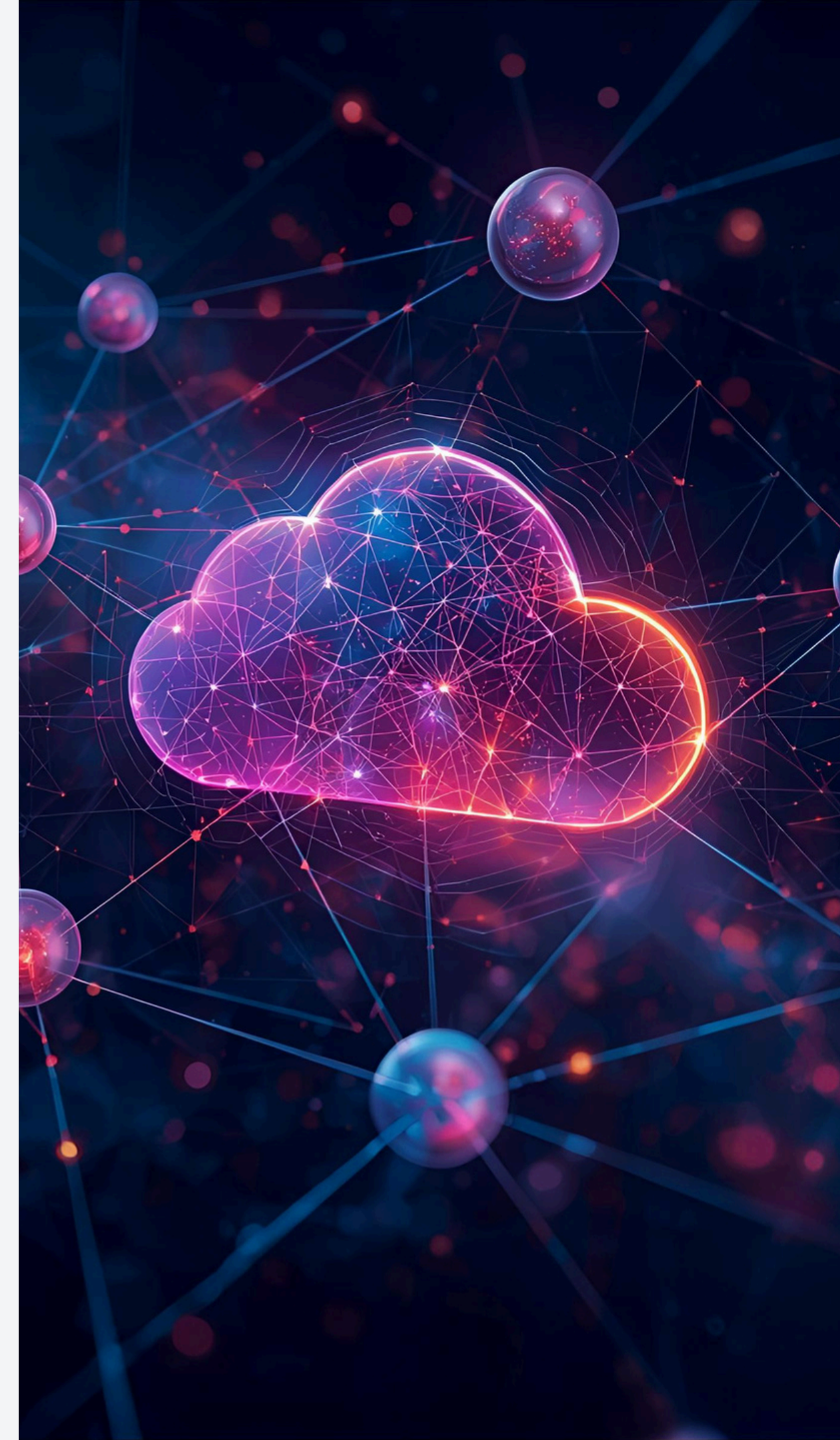
01 / PART

Federated IoT Vision

Federated IoT Vision

From Cloud to Edge Federation

Federated IoT moves anomaly detection from centralized clouds to cooperative edge nodes that share model updates, not raw data. This enables privacy-preserving, low-latency threat recognition inside bandwidth-starved micro-sensor fields while keeping sensitive readings on device.



Smart Dust Covert Mesh

Millimeter-scale, battery-free motes form self-healing meshes. They harvest ambient energy, hide in plain sight, and monitor for vibration, chemicals, or motion without revealing their presence, making traditional centralized detection architectures impossible.

- 🌱 **Energy Autonomy: Harvests from vibration & RF.**
- 🔗 **Covert Mesh: Self-healing & virtually undetectable.**





02 / PART

Federated Detection

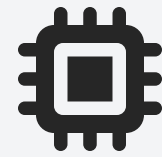
On-Device TinyML Pipeline

Each mote runs efficient models on ultra-low-power MCUs, compressing data into tiny gradients for selective, event-driven transmission.



Sensor Trace

24h Data



Quantized Model

< 1 mW



Tiny Gradient

128 Bytes

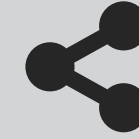


LoRa Send

On Risk

Gradient Consensus Protocol

Neighbors perform sparse gossip averaging with reputation-weighted aggregation. Updates that deviate significantly from the community norm are rejected, preventing poisoned models from propagating.



**Gossip
Averaging**



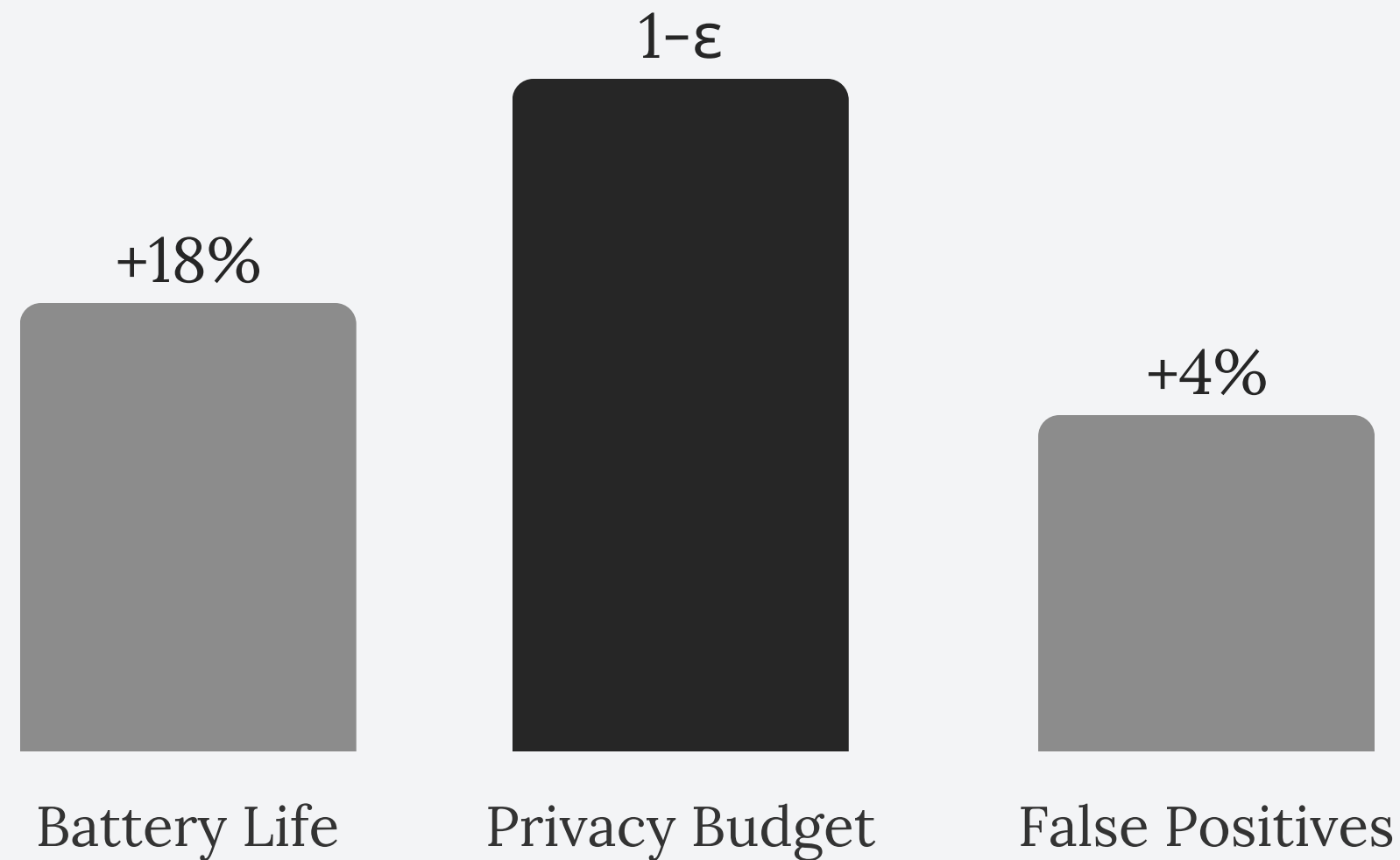
**Reputation
Weight**



**Reject
Outliers**

Privacy & Energy Trade-offs

Balancing stealth, power, and accuracy with adaptive differential privacy.



Adaptive algorithms optimize the trade-off based on available solar energy.



03 / PART

Covert Deployment

Urban Camouflage

Motes are designed to be visually and electronically invisible, blending into the urban environment to avoid detection.

-  Disguised as screws, bird droppings, or dust.
-  Photovoltaic paint matches surrounding albedo.
-  Optical sync below 0.1 lux to evade NVGs.

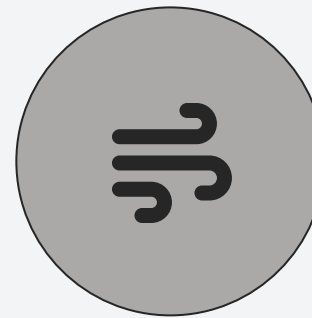


Self-Healing Mesh Tactics

When nodes are lost, the network autonomously adapts to restore coverage within minutes.



Boost Power
+3 dB TX



Reroute
Via HVAC



Opportunistic Relay
Idle BT Devices

95% Coverage Restored in 90 Minutes



04 / PART

Security Threats

Model Poisoning & Evasion

Adversaries can attempt to fool the system by injecting bad data or altering the environment. The network defends itself through community consensus and local outlier detection.

Attack: Malicious Gradients

Classifying gunshots as traffic noise.

Attack: Environmental Shifts

Spraying paint to alter vibration baselines.



Defense: Reputation & Rollback

The system detects and neutralizes attacks within three gossip rounds using reputation scoring and local outlier storage to rollback malicious changes.

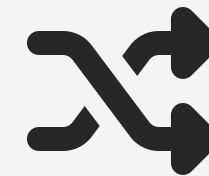
RF Side-Channel Leakage

Continuous transmission risks detection via direction-finding. A stealth protocol is used to minimize the chance of interception.



Continuous Chirp

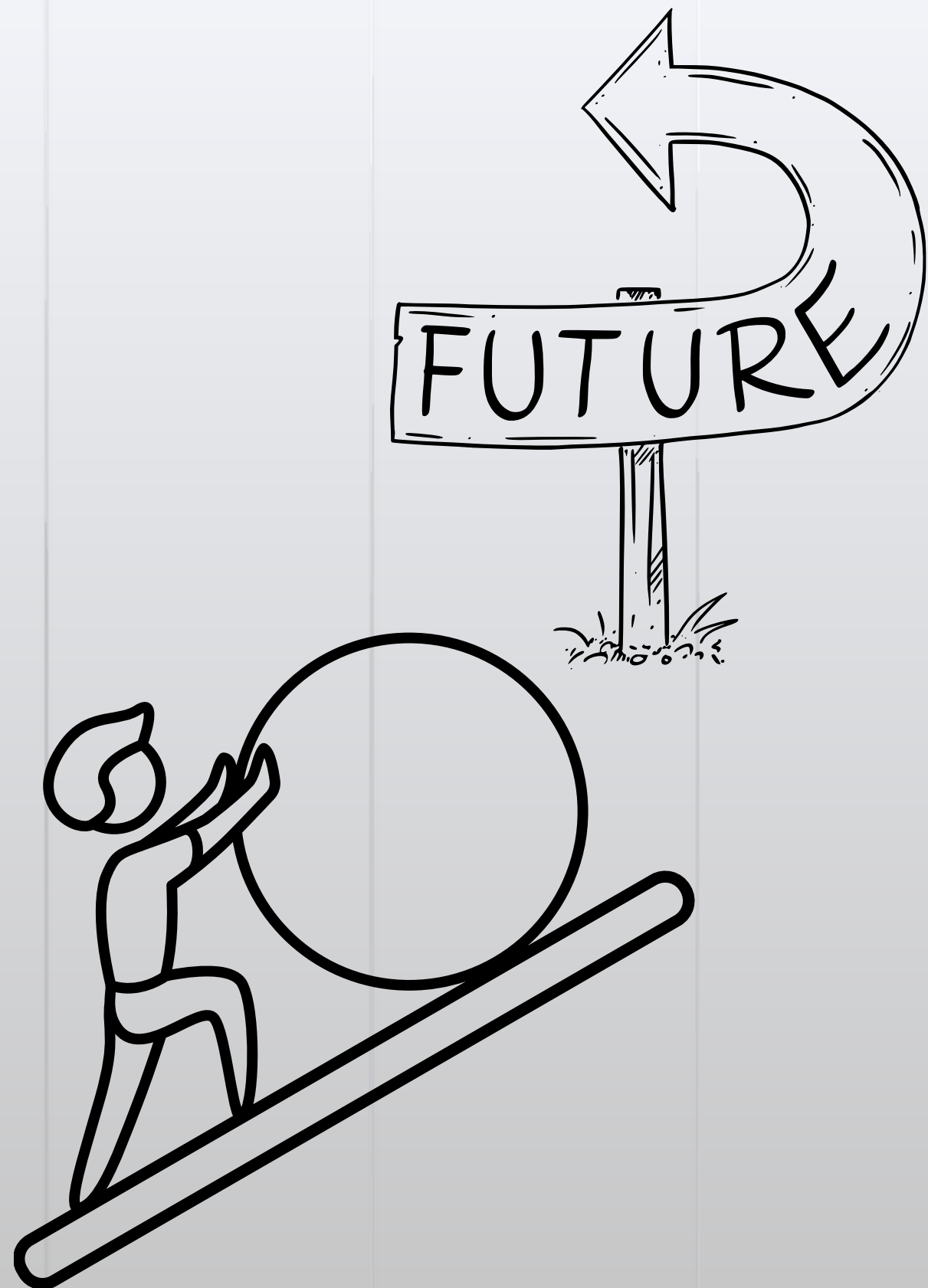
High Risk



Stealth Protocol

Low Risk

Result: < 5% Intercept Probability @ 100m with an average current draw of only < 50 μ A.



05 / PART

Resilience & Future

Blockchain Audit Anchors

To ensure integrity without compromising stealth, a tamper-evident audit trail is created using an energy-harvesting NFC tag.



Daily Merkle Root

A hash of all model checksums is computed and stored.



NFC Tag Etching

The root hash is physically written to a passive NFC tag.



Smartphone Scanning

An investigator can later scan the tag to verify firmware history.



Post-Quantum Micro-Crypto

Future-proofing security with Dilithium signatures, optimized for the extreme constraints of Smart Dust.

Dilithium Sig

1.2kB



Fragments

5x256B



Overhead

+0.8%

Provides 128-bit post-quantum authenticity with minimal energy impact on a Cortex-M0+.

Toward Invisible Sentinels

The next evolution is ephemeral. Biodegradable smart dust will dissolve after its mission, leaving no trace. This enables powerful, ethical monitoring for disaster response or supply-chain integrity without the risk of permanent surveillance.

Biodegradable

Ephemeral

Ethical Monitoring





THANK YOU



Regards @MajorKali



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